Day 20 Task: Linux User Management

Linux User Management

Linux user management is the process of creating, modifying, organizing, and removing user accounts on a Linux operating system. It helps administrators control who can access the system and what actions each user is permitted to perform.

Linux provides user accounts, groups, ownership, and access permissions to protect system resources. Proper user management improves security, supports accountability, and prevents unauthorized access.

Linux User Accounts

A user account allows a person or service to access and use a Linux system. Each account is associated with a unique User ID (UID), a username, and other account information.

Linux commonly uses the following account types:

Root User: The root account is the superuser with extensive administrative privileges. It can perform system-wide operations, modify protected files, and manage other accounts.

Regular User: A regular user has limited permissions and can generally access personal files and perform permitted tasks. Administrative operations usually require authorized privilege escalation.

System User: System accounts are typically used by services and applications to run processes with specific permissions rather than unrestricted access.

Linux Groups

A Linux group is a collection of users that can share access permissions to files, directories, and other resources. Each group has a unique Group ID (GID).

Groups simplify permission management because administrators can grant access to a group instead of assigning the same permissions to each user individually.

For example, a developers group may be allowed to access a shared project directory, while other users are denied access.

Useful commands include:

groups

Displays the groups associated with the current user.

id username

Displays the UID, primary GID, and supplementary groups for a user.

sudo groupadd developers

Creates a group named developers.

Creating Linux Users

The useradd and adduser commands can be used to create user accounts, depending on the Linux distribution.

sudo useradd -m alice

Creates a user named alice and creates a home directory for the account.

sudo passwd alice

Sets or changes the password for alice.

The home directory stores a user's personal files and configuration settings. Creating separate accounts allows users to work independently without sharing the same identity or personal files.

Modifying User Accounts

Administrators may need to modify user accounts when responsibilities change or when users require access to additional resources.

sudo usermod -aG developers alice

Adds alice to the supplementary developers group. The -aG options append the group without replacing existing supplementary group memberships.

sudo usermod -s /bin/bash alice

Changes the login shell for alice to /bin/bash, assuming that shell is installed and permitted.

sudo passwd -l alice

Locks password-based authentication for the account on systems that support this behavior. This does not necessarily disable every possible authentication method or terminate existing sessions.

Deleting User Accounts

When an account is no longer needed, administrators should disable or remove it according to organizational policy.

sudo userdel alice

Deletes the user account but generally leaves the user's home directory and files behind.

sudo userdel -r alice

Deletes the account and attempts to remove its home directory and mail spool. This operation can permanently remove data, so it should only be performed after confirming that the files are no longer needed.

Before deleting an account, administrators should check file ownership, running processes, scheduled jobs, and access to shared resources.

Linux User and Group Files

Linux stores local account and group information in specific system files.

/etc/passwd contains basic account information, including usernames, UIDs, primary GIDs, home directories, and login shells. Password hashes are generally not stored in this file.

/etc/shadow stores password hashes and password-aging information on systems that use shadow passwords. Access is restricted because the information is security-sensitive.

/etc/group contains group definitions and supplementary group membership information.

/etc/gshadow stores protected group information on systems that use it.

These files are important for account administration and security. Unauthorized modifications can disrupt logins or compromise system access.

File Ownership and Access Permissions

Linux uses file ownership and permissions to determine which users can access files and directories.

The ls -l command displays permissions and ownership:

ls -l

The output identifies the file owner, associated group, and permission settings.

The chown command changes file ownership:

sudo chown alice:developers project.txt

This assigns alice as the owner and developers as the associated group.

The chmod command changes file permissions:

chmod 640 project.txt

This gives the owner read and write permissions, the group read permission, and others no permissions.

Combining appropriate user accounts, groups, and file permissions helps ensure that only authorized users can access sensitive resources.

Why Separate User Accounts Are Important

Separate user accounts improve security by giving each person an individual identity and appropriate permissions.

They help administrators track activity, restrict access to personal files, prevent users from unintentionally modifying one another's work, and investigate security incidents.

Shared accounts should be avoided where possible because they make it more difficult to determine which individual performed a particular action.

Restricting Root Access

The root account has extensive privileges, so unauthorized or accidental use can cause significant damage to a system.

Linux administrators should use regular accounts for everyday activities and use sudo for authorized administrative tasks. Access to sudo should be restricted to trusted users and reviewed regularly.

Multifactor authentication, strong passwords, security updates, and monitoring of privileged activity provide additional protection.

Principle of Least Privilege

The principle of least privilege means that each user, process, and application should receive only the permissions necessary to perform its required tasks.

For example, a user who only needs to read a report should not receive permission to modify or delete it. Similarly, a web server process should not run with unrestricted root privileges unless absolutely necessary.

Applying this principle reduces the potential damage caused by mistakes, compromised accounts, or malicious software.